Standard Correction

LLM Labels $\hat{Z}$
error rates vary across subgroups

Pooled Confusion Matrix
one $\bar{C}$ for all subgroups

Uniform Inversion
$\bar{C}^{-1}$ applied identically

Bias Amplification
over-corrects low-error, under-corrects high-error

EC-HTE (Ours)

LLM Labels $\hat{Z}$ + Expert Sample
stratified validation labels

Hierarchical Bayes
per-subgroup $\hat{C}_s$ with partial pooling

Stratified Inversion + κ-guard
$\hat{C}_s^{-1}$ tailored to each subgroup

Corrected CATE + CI
bias reduced, uncertainty propagated